

Lab 02 — Testing a Three-Branch Classifier

Module: COMP H4204 — Games Engineering II (Software Testing)

1. Objectives

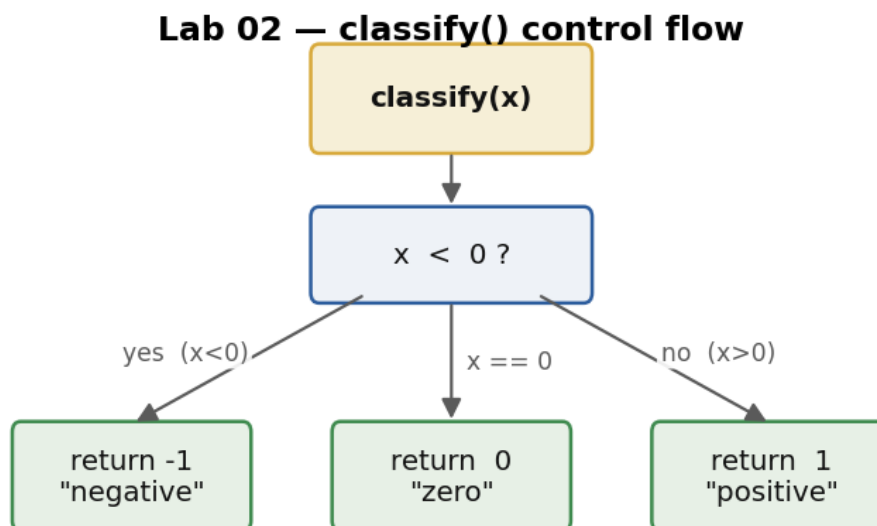
By the end of this lab you will be able to:

1. Recognise a *branch* in a small if / else if / else program.
2. Write a test case as an *expected value* compared against the *actual return value* of a function.
3. Count branch coverage by hand.

2. Learning outcomes

- **LO1** — Explain the difference between a test *input* and a test *oracle* (the expected value).
- **LO2** — Enumerate the branches of a three-way decision.
- **LO3** — Design the minimum set of tests that visits every branch at least once.

3. Diagram



Three branches → three tests achieve 100 % branch coverage

classify() control flow

4. Problem statement

You are given a one-file program lab02_classify.cpp containing:

- A function `int classify(int x)` that returns -1 when x is negative, 0 when x is zero, and 1 when x is positive.
- An empty `runTests()` function where you will place your tests.

Your job is to add three test cases inside `runTests()` — one per branch — using the tiny test helper (`checkEqual`) that is already provided. No external framework is needed this week.

5. Tasks

1. Copy the starter into `studentLabs/lab02/`.
2. Read `classify(x)` and identify the three branches (negative, zero, positive).
3. Inside `runTests()`, add three calls to `checkEqual(actual, expected, "label")`, one for each branch.
4. Compile and run:
5. All three tests should print PASS. The summary line at the end should read Summary: 3 passed, 0 failed OR, x passed , x failed (depending upon the values that you provide it).
6. Extend the code by providing user input to check branch coverage for all three scenarios. The program must quit when user enters any character.

6. Standard submission

Every lab in this module uses the same submission format. Place the following in `studentLabs/lab02/`:

1. **Updated source code** — `lab02_classify.cpp` with your three tests filled in.
2. **VSCode screenshot** — `screenshots/vscode.png`.
3. **Terminal screenshot** — `screenshots/terminal.png`.
4. **Lab report** — Briefly explain why complete branch coverage is required?